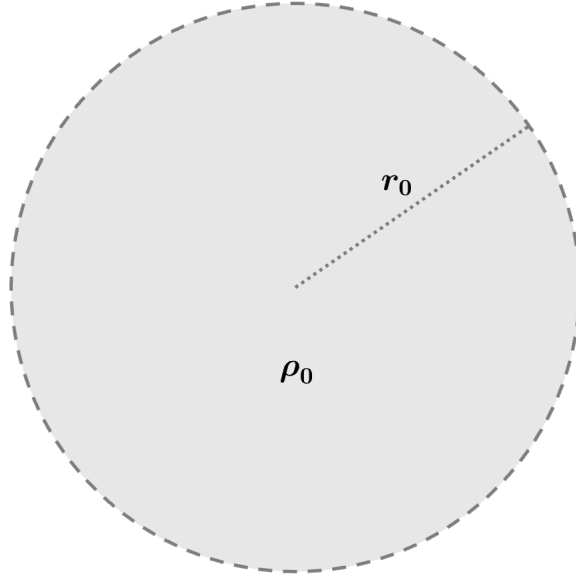


2013 F=ma Exam: Problem 16

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We can use dimensional analysis to find how the time T of collapse scales with ρ_0 , r_0 , and G :

$$T \propto \rho_0^\alpha r_0^\beta G^\delta$$

We first find the dimensions of each quantity:

$$[T] = T$$

$$[\rho_0] = \frac{M}{L^3}$$

$$[r_0] = L$$

To find the dimensions of G , we use

$$[F] = \frac{[G][m_1][m_2]}{[r]^2}$$

$$\frac{ML}{T^2} = \frac{[G]M^2}{L^2}$$

so

$$[G] = \frac{L^3}{MT^2}$$

We take the dimensions of the first equation,

$$[T] = [\rho_0]^\alpha [r_0]^\beta [G]^\delta$$

$$T = \left(\frac{M}{L^3}\right)^\alpha L^\beta \left(\frac{L^3}{MT^2}\right)^\delta$$

Counting powers of time T ,

$$1 = -2\delta$$

$$\delta = -\frac{1}{2}$$

Counting powers of mass M ,

$$0 = \alpha - \delta$$

$$\alpha = \delta = -\frac{1}{2}$$

Counting powers of length L ,

$$0 = -3\alpha + \beta + 3\delta$$

$$\beta = 3\alpha - 3\delta = 0$$

Thus,

$$T \propto \frac{1}{\sqrt{G\rho_0}}$$

so the answer is D.